Lecture #28 Essentials of Genetics

ADVANTAGES AND LIMITATIONS OF PCR

ADVANTAGES

- Specific
- Simple, rapid, relatively inexpensive
- Amplifies from low quantities
- Works on damaged DNA
- Sensitive
- Flexible

LIMITATIONS

- Contamination risk
- Primer complexities
- Primer-binding site complexities
- Amplifies rare species
- Detection methods

Lecture #29-Essentials of Genetics

APPLICATIONS OF PCR

APPLICATIONS

- Detection of Infectious diseases: AIDS, TB, CMV, H1N1, etc Viral, Bacterial and fungal infections,
- Diagnosis of latent viruses.
- Forensic applications: DNA finger printing

APPLICATIONS

- Detection of Mutations: Inherited disorders & carriers Track DNA abnormalities
- Prenatal diagnosis of genetic disorders.

Applications of PCR

APPLICATIONS

- Detection of pathogens Pre-natal diagnosis DNA fingerprinting Gene therapy Mutation screening
- Drug discovery
- Classification of organisms Genotyping

APPLICATIONS

- Molecular Archaeology
- Molecular Epidemiology
- Molecular Ecology
- Bioinformatics
- Genomic cloning
- Site-directed mutagenesis
- Gene expression studies

Applications of PCR

CLINICAL MICROBIOLY

- Identification of bacteria:
- Slow growing bacteria
- Bacterial antibiotic resistance genes.
- VIRUS: almost all viruses identification and viral load.

CLINICAL MICROBIOLY

- FUNGUS:
- Aspergillus spp
- Candida spp
- Plasmodium spp
- Trypanosoma spp
- Leishmania spp
- Babesia spp

Lecture #30 Polymerase chain reaction (PCR)

Applications of PCR

- PCR has widespread applications in various fields of life sciences including genetic engineering, medical, forensic, agriculture, environment etc.

PCR-Gene cloning and expression

- PCR has been used in gene cloning and screening of genomic libraries

Lecture #31- Polymerase chain reaction (PCR)

PCR-Medicine

- PCR has major impact on medicine especially in the field of clinical microbiology or diagnosis
- Molecular tools have also allowed to perform prenatal genetic diagnosis

Lecture #32 Polymerase chain reaction (PCR)

PCR-Forensic sciences

- Forensic science is the application of scientific procedures to solve criminal and legal matters
- Molecular methods are used to established the filiations of a person or to obtain evidence from minimal samples of saliva, semen or other tissues

PCR-DNA profiling

- DNA profiling or DNA fingerprinting is a forensic technique used to identify individuals by characteristics of their DNA

Lecture #33 Polymerase chain reaction (PCR)

PCR-Agricultural sciences and environment

- PCR has also facilitated research in detection of pathogens in plants, animals and environment

Lecture #34 Polymerase chain reaction (PCR)

PCR-Molecular paleontology

- **Molecular paleontology refers to the recovery and analysis of DNA and protein from ancient human, animal and plant remains**

Lecture #35- Cutting DNA molecules

Types of restriction and modification (R-M) system

- At least four R-M systems are known
- Type I
- Type II
- Type III
- Type IIs

Type I

- Type I systems were the first to be characterized from *E. coli* K12
- The active enzyme consists of two restriction subunit, two modification subunit and one recognition subunit
- Type I systems are of little value for gene manipulation

Cutting DNA molecules

Type II

- Most of the useful R-M system is Type II
- Type II enzymes recognize a defined sequence and cut within it

Type III

- Type III enzymes have symmetrical recognition sequences but otherwise resemble type I systems and are of little value

Cutting DNA molecules

Type IIs

- Type IIs systems have similar cofactors and structure to type II but restriction occurs at a distance from recognition site that limits their usefulness

Lecture #36- Cutting DNA molecules

Nomenclature

- A suitable system was proposed by Smith and Nathans (1973)
- The species name of host organisms is identified by the first letter of genus and first two letters of specific epithet
- *E. coli* = Eco
- *H. influenzae* = Hin

Nomenclature

- Strain identification is written as *Eco*K
- In case, host strain has several restriction and modification systems, these are identified by roman numerals, for example, in case of *H. influenzae*
- *HindI, HindII, HindIII* etc

Cutting DNA molecules

Nomenclature

- **All restriction enzymes have general name endonuclease R and modification-methylase M followed by the system name, for example, in case of *H. influenzae***
- **R. *HindIII* or M. *HindIII***

Examples of restriction endonuclease nomenclature

Enzyme	Enzyme source	Recognition sequence
SmaI	***Serratia marcescens***, 1st enzyme	**CCCGGG**
HaeIII	***Haemophilus aegyptius***, 3rd enzyme	**GGCC**
HindII	***H. influenzae***, strain d, 2nd enzyme	**GTPyPuAC**
HindIII	***H. influenzae***, strain d, 3rd enzyme	**AAGCTT**
HamHI	***Bacillus amyloliquefaciens***, strain H, 1st enzyme	**GGATCC**

Lecture #37- Cutting DNA molecules

Target sites

- **Type II endonucleases recognize and cleave DNA within particular sequences of four to eight nucleotides which have a twofold axis of rotational symmetry i.e. referred as *palindromes***

5′-GAATTC-3′

5′-CTTAAG-3′

5′-G A A T T C-3′

3′-C T T A A G-5′

5′-G/ A A* T T C-3′

3′-C T T A* A/ G-5′

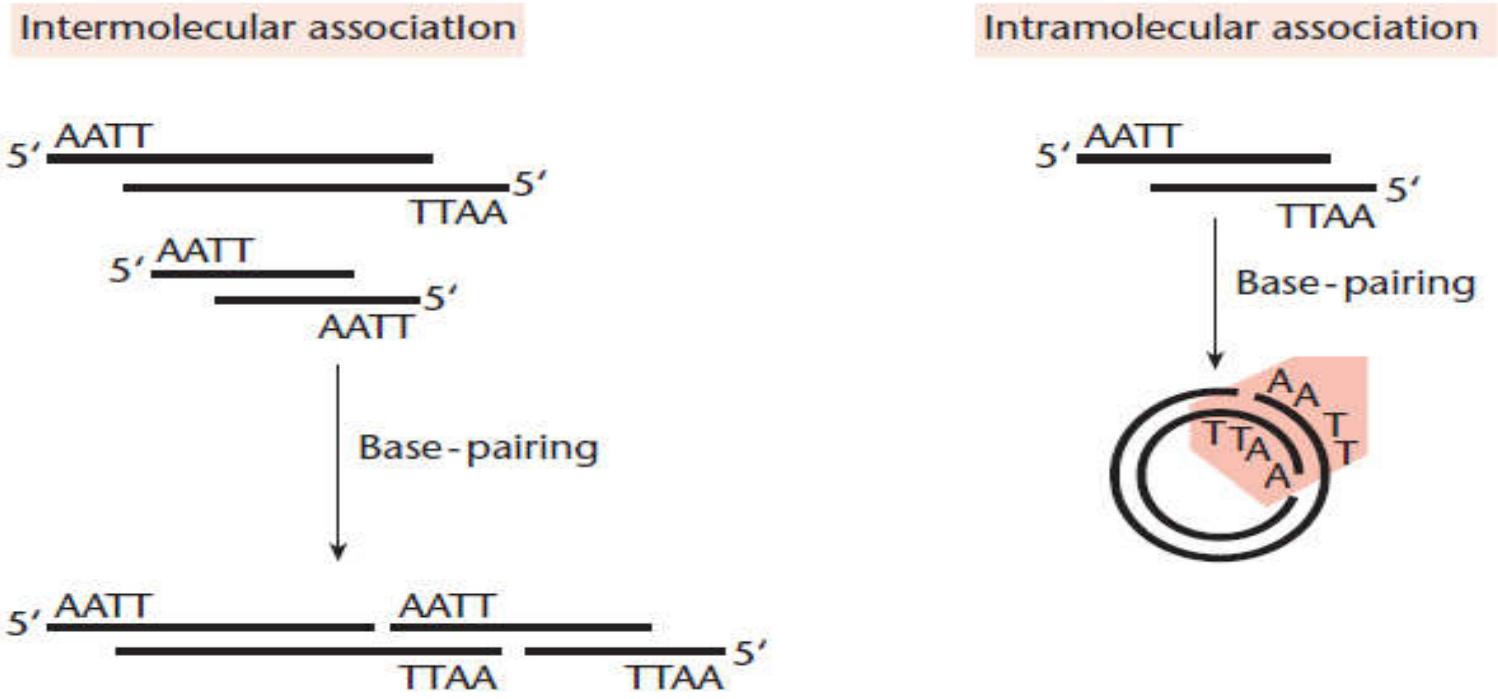
5′-G

3′-C T T A A-5′

5′- A A T T C-3′

G-5′

Single stranded breaks by *Eco*R1



Cohesive fragments of DNA produced by digestion with EcoR1

Lecture #38- Cutting DNA molecules

Number and size of restriction fragments

- The number and size of fragments generated by restriction enzyme depend on the frequency of occurrence of the target site in the DNA to be cut

Number and size of restriction fragments

- Four base recognition site occurs every 4^4 (256) bp
- Six base recognition site occurs every 4^6 (4096) bp
- Eight base recognition site occurs 4^8 (65,536) bp

Average fragment size (bp) produced by different enzymes

Enzyme	Target	Arabidopsis	E. Coli	Human
*Apa*I	GGGCCC	25000	15000	2000
*Bam*HI	GGATCC	6000	5000	5000
*Spe*I	ACTAGT	8000	60000	10000

Lecture #39- Cutting DNA molecules

Summary of restriction endonucleases

- **Type II restriction enzymes are heavily responsible for the current explosion in the field of gene manipulation in that they are essential in forming recombinant DNA molecules**

Lecture #40- Joining DNA molecules

DNA modifying enzymes

- Nucleases
- DNA Polymerase
- Reverse transcriptases
- DNA ligases

Nucleases

- Nucleases or DNases are the enzymes that degrade DNA
- Two broad classes of nucleases
    - i). Exonucleases
    - ii). Endonucleases

Joining DNA molecules

DNA Polymerase

- An enzyme that catalyzes template-dependent synthesis of DNA

Reverse transcriptases

- An RNA-directed DNA polymerase in retroviruses; capable of making DNA complementary to an RNA

Joining DNA molecules

DNA ligase

- **An enzyme that creates a phosphodiester bond between the 3′ end of one DNA segment and the 5′ of another**

Lecture #41- Joining DNA molecules

Methods of joining DNA fragments

- **Mainly three methods are used for joining DNA in vitro**

i). Joining covalently annealed cohesive ends by DNA ligase

ii). Joining blunt-ended fragments by DNA ligase from phage T4 infected *E. coli*

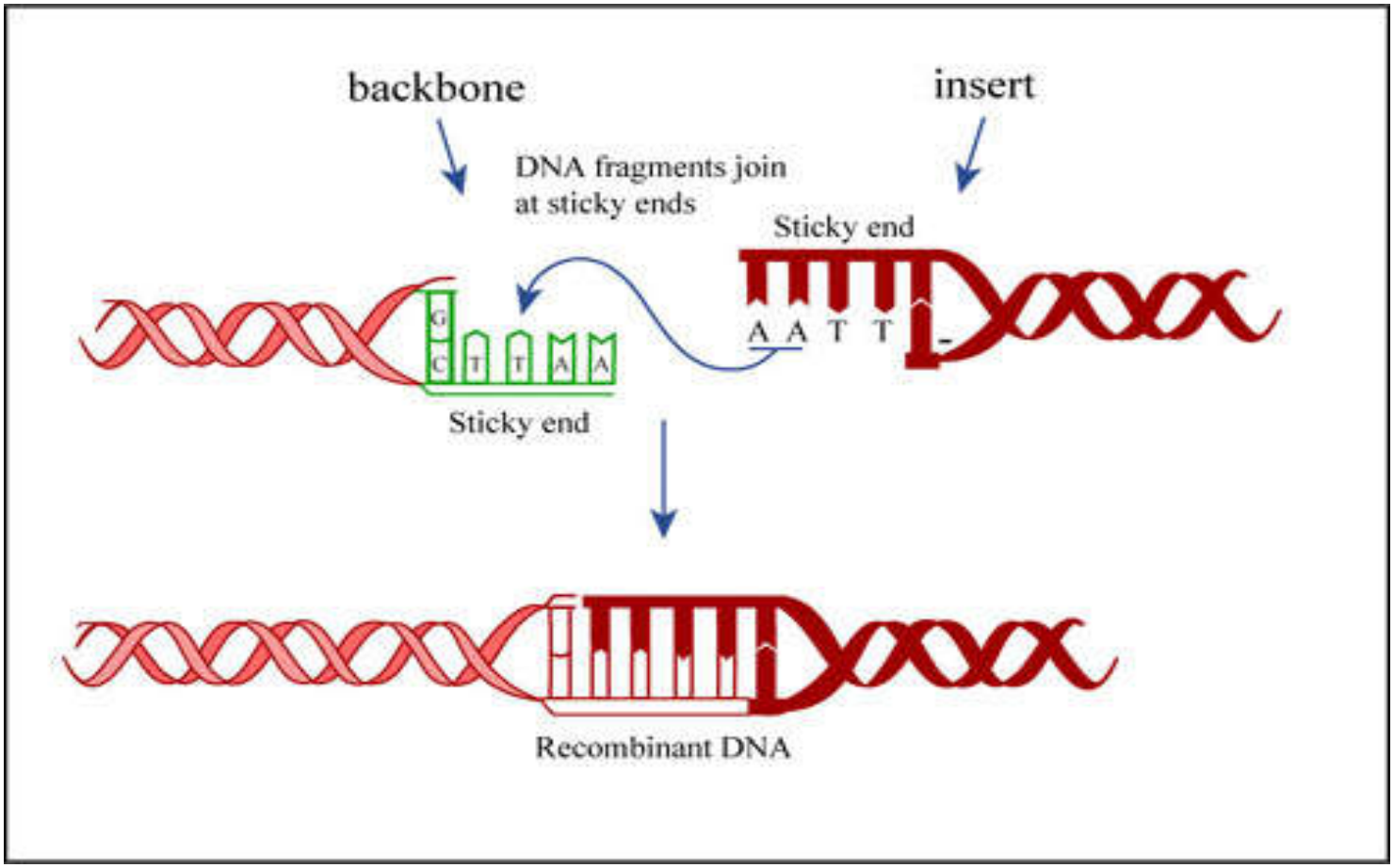
Methods of joining DNA fragments

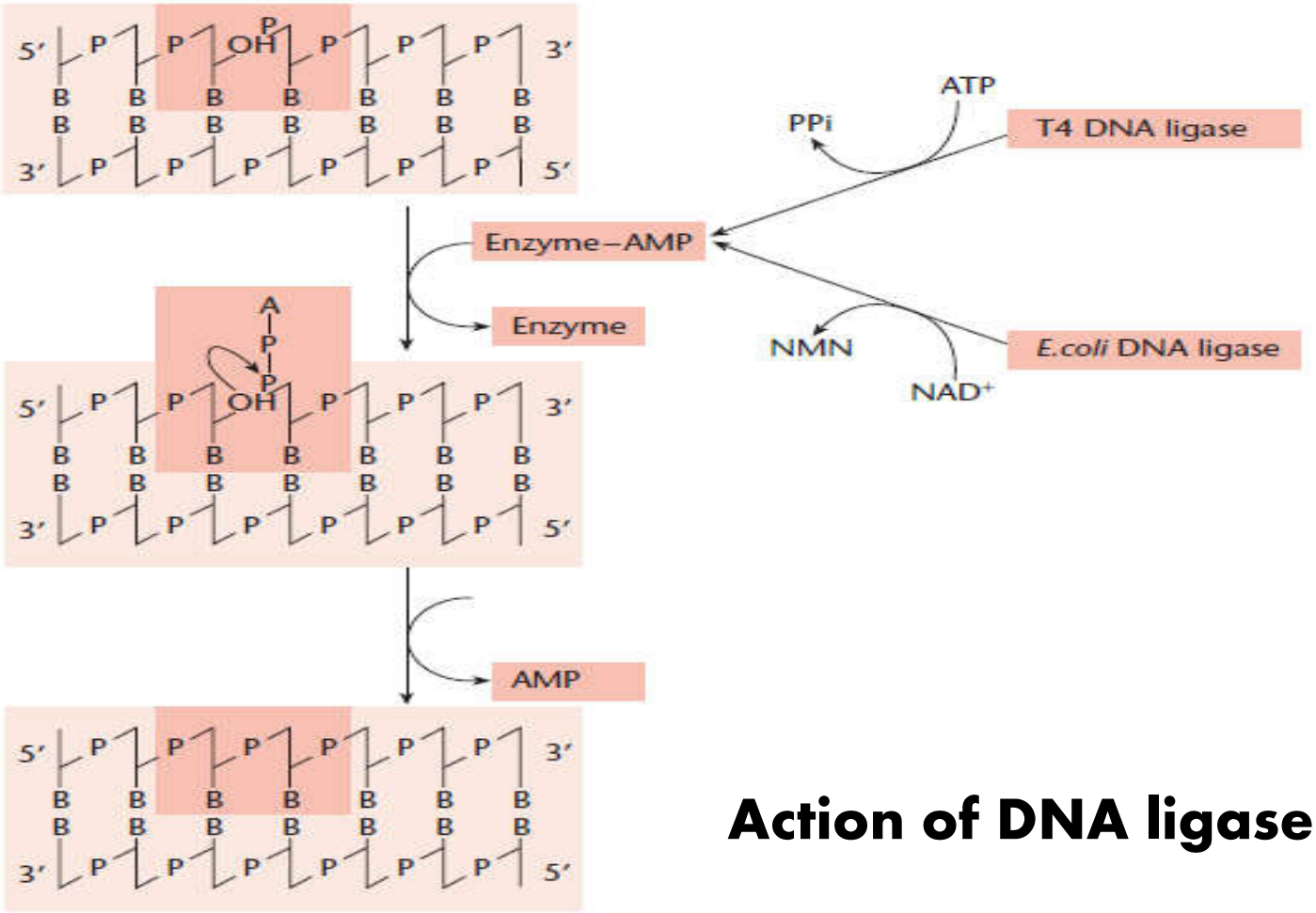
iii). The third utilizes the enzyme terminal deoxynucleotidyltransferase to synthesize homopolymeric 3′ single stranded tails at the ends of fragments

Joining DNA molecules

DNA ligase

- ***E. coli*** **and phage T4 encode enzyme, DNA ligase, which seals single stranded nicks between adjacent nucleotides in a duplex DNA chain**


backbone
insert
DNA fragments join
at sticky ends
Sticky end
G
C T T A A
Sticky end
A A T T
Recombinant DNA



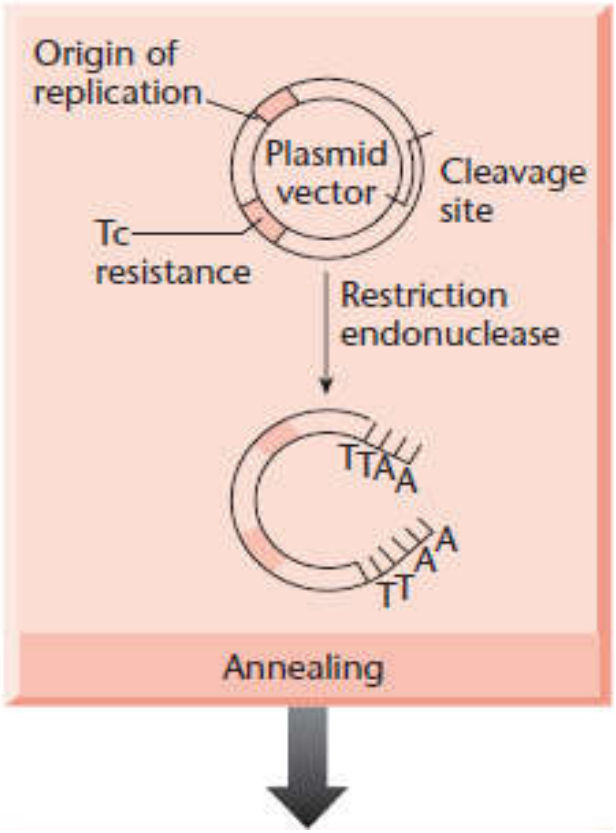
Action of DNA ligase

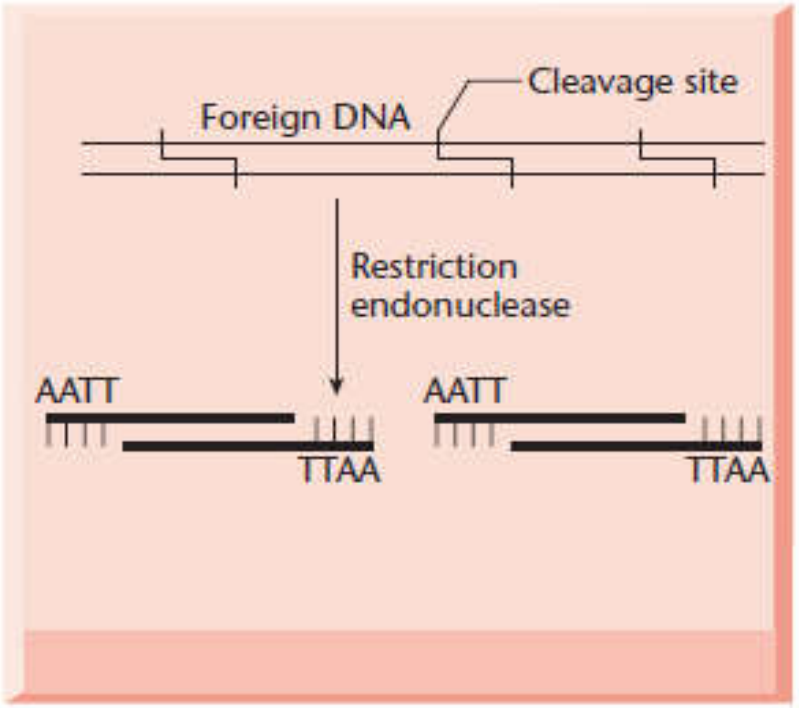
Lecture #42-Joining DNA molecules

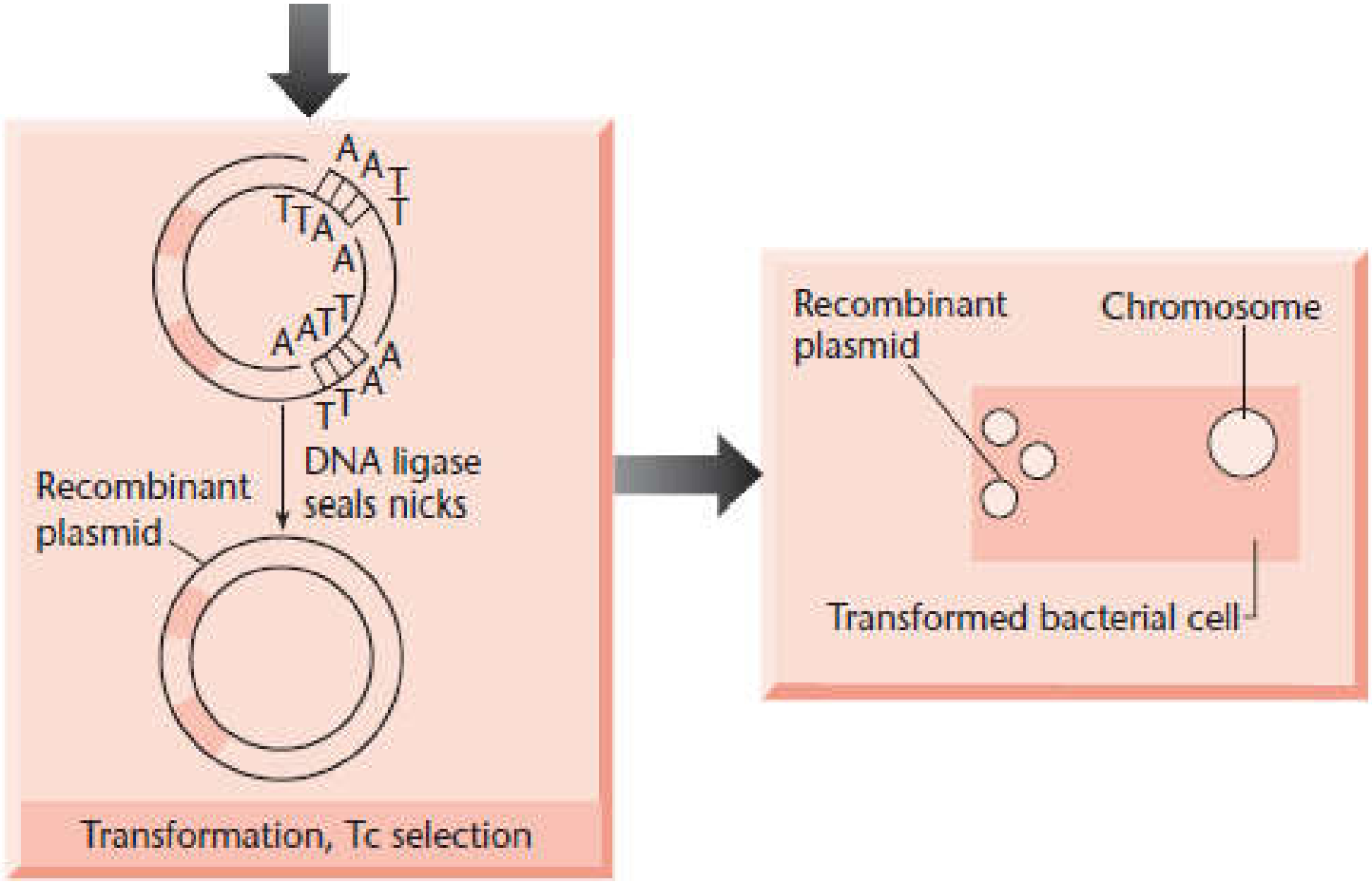
DNA ligase to create covalent recombinant DNA

- **DNA ligase can repair the nicks produced after the association of cohesive ends of DNA strands**

Use of DNA ligase to create a covalent DNA recombinant





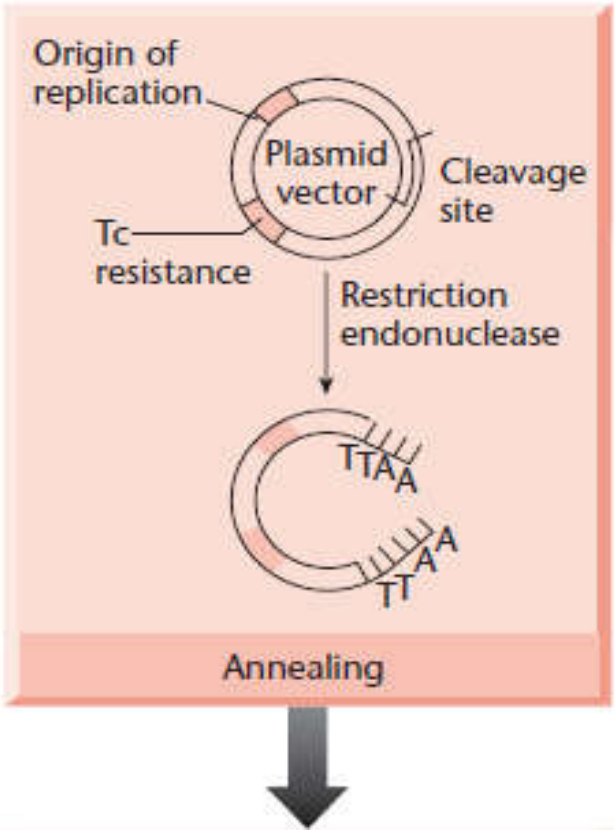

DNA ligase
seals nicks
Recombinant
plasmid
Transformation, Tc selection
Recombinant
plasmid
Chromosome
Transformed bacterial cell

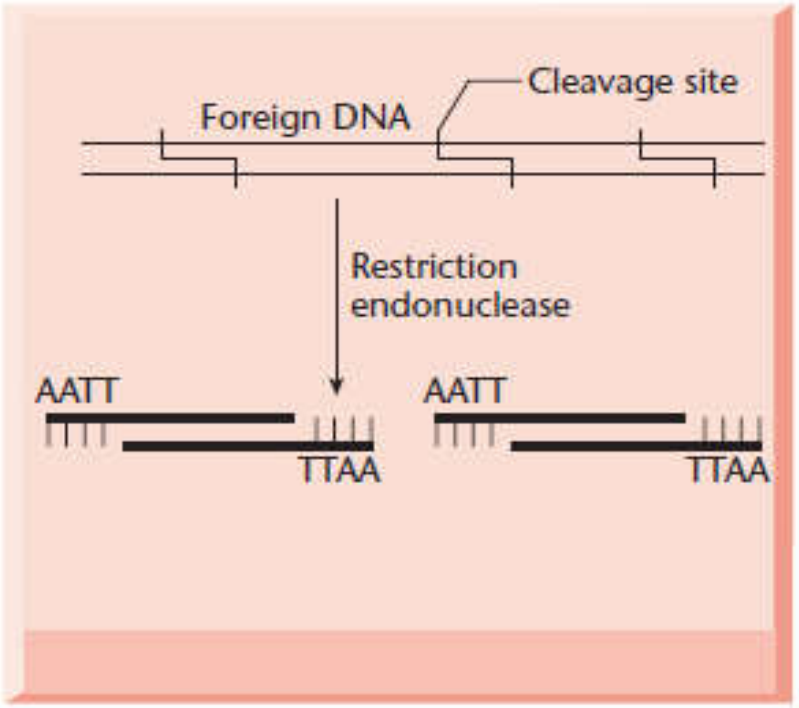
Lecture #43- Joining DNA molecules

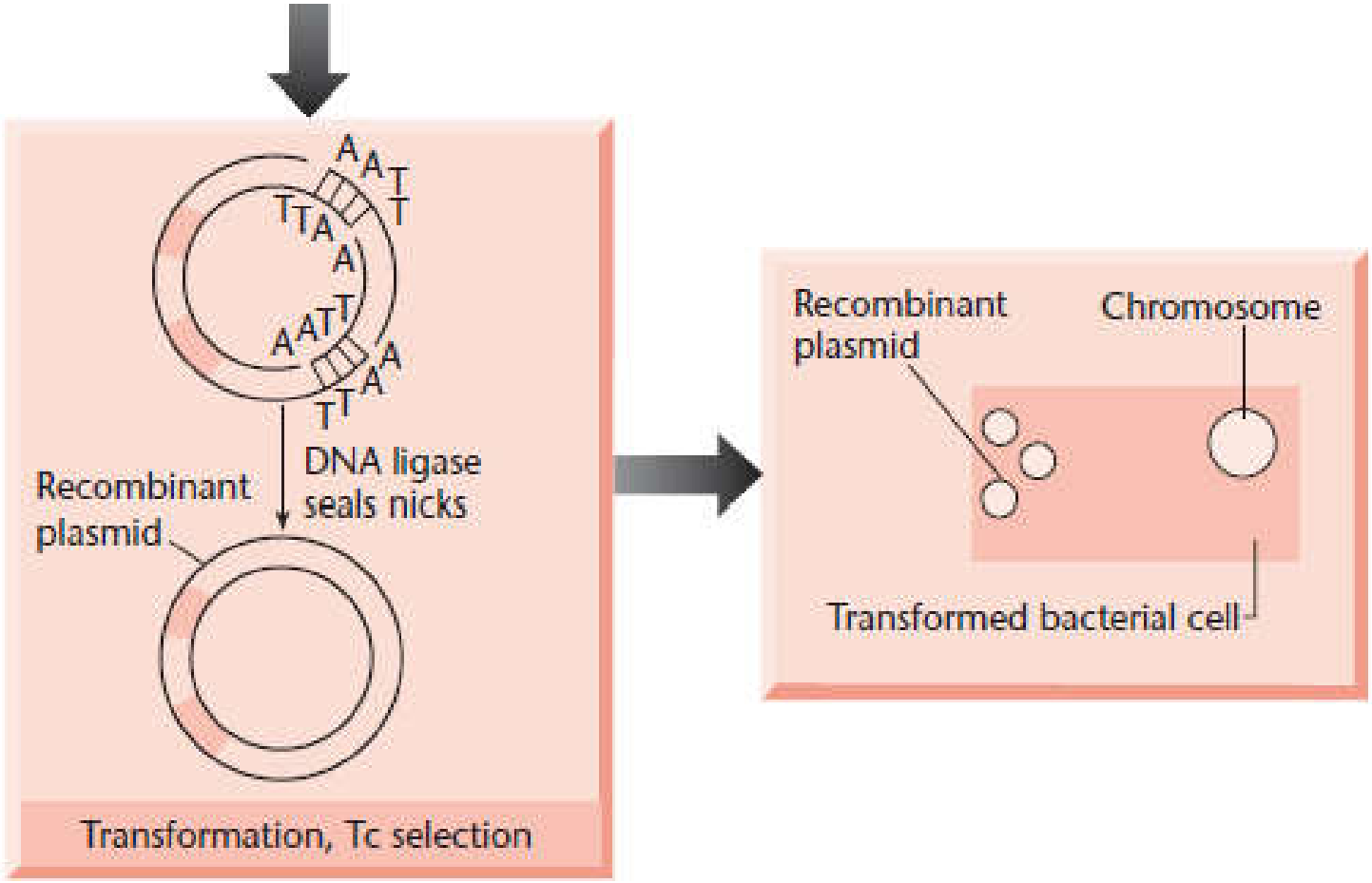
DNA ligase to create covalent recombinant DNA

- **DNA ligase can repair the nicks produced after the association of cohesive ends of DNA strands**

Use of DNA ligase to create a covalent DNA recombinant






DNA ligase
seals nicks
Recombinant
plasmid
Transformation, Tc selection
Recombinant
plasmid
Chromosome
Transformed bacterial cell

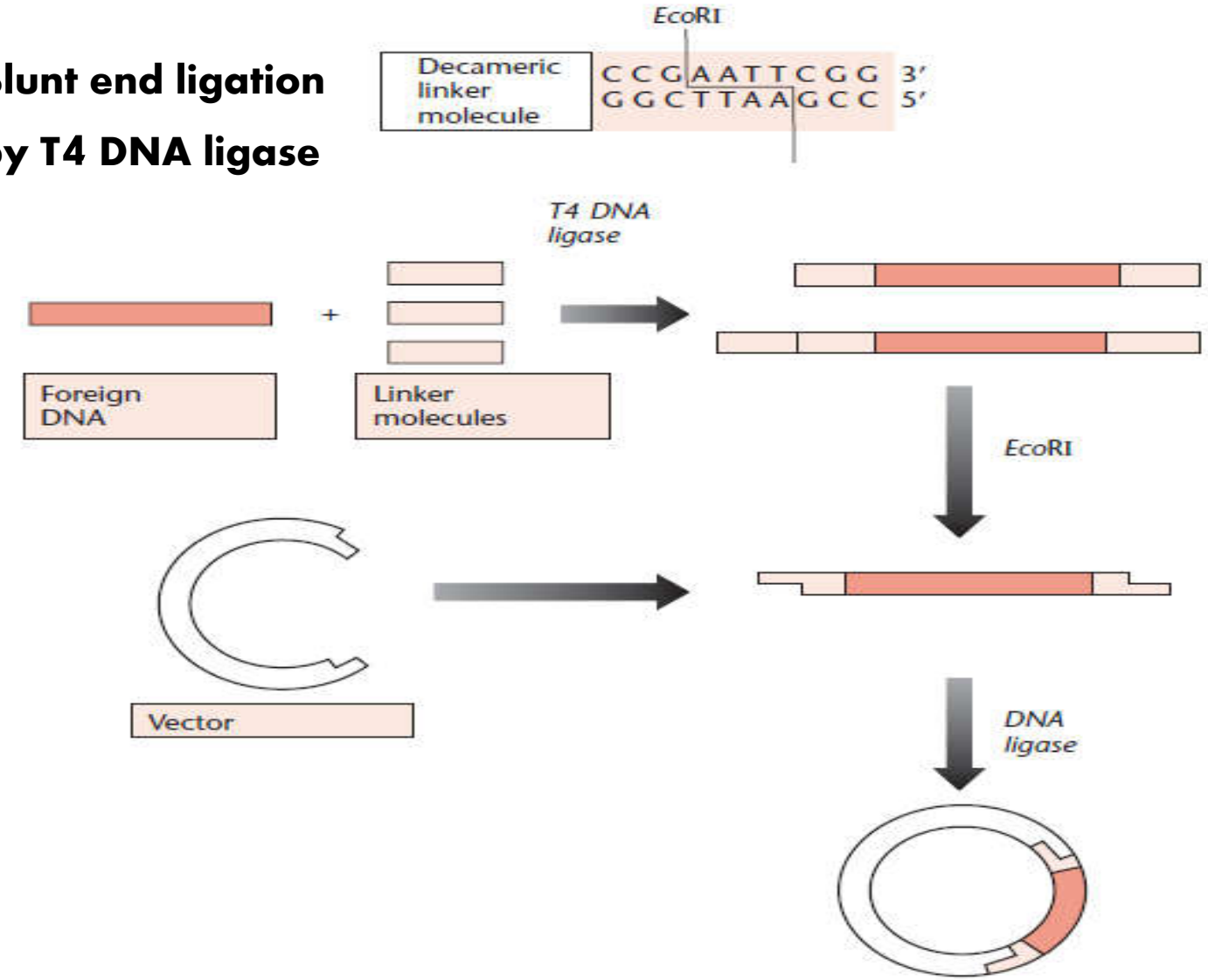
Lecture #44 -Joining DNA molecules

Blunt end ligation via linker molecules

- **Joining DNA fragments with cohesive ends by DNA ligase is a relatively efficient process**
- **T4 DNA ligase has been used to joint blunt-ended DNA molecules**

Blunt end ligation

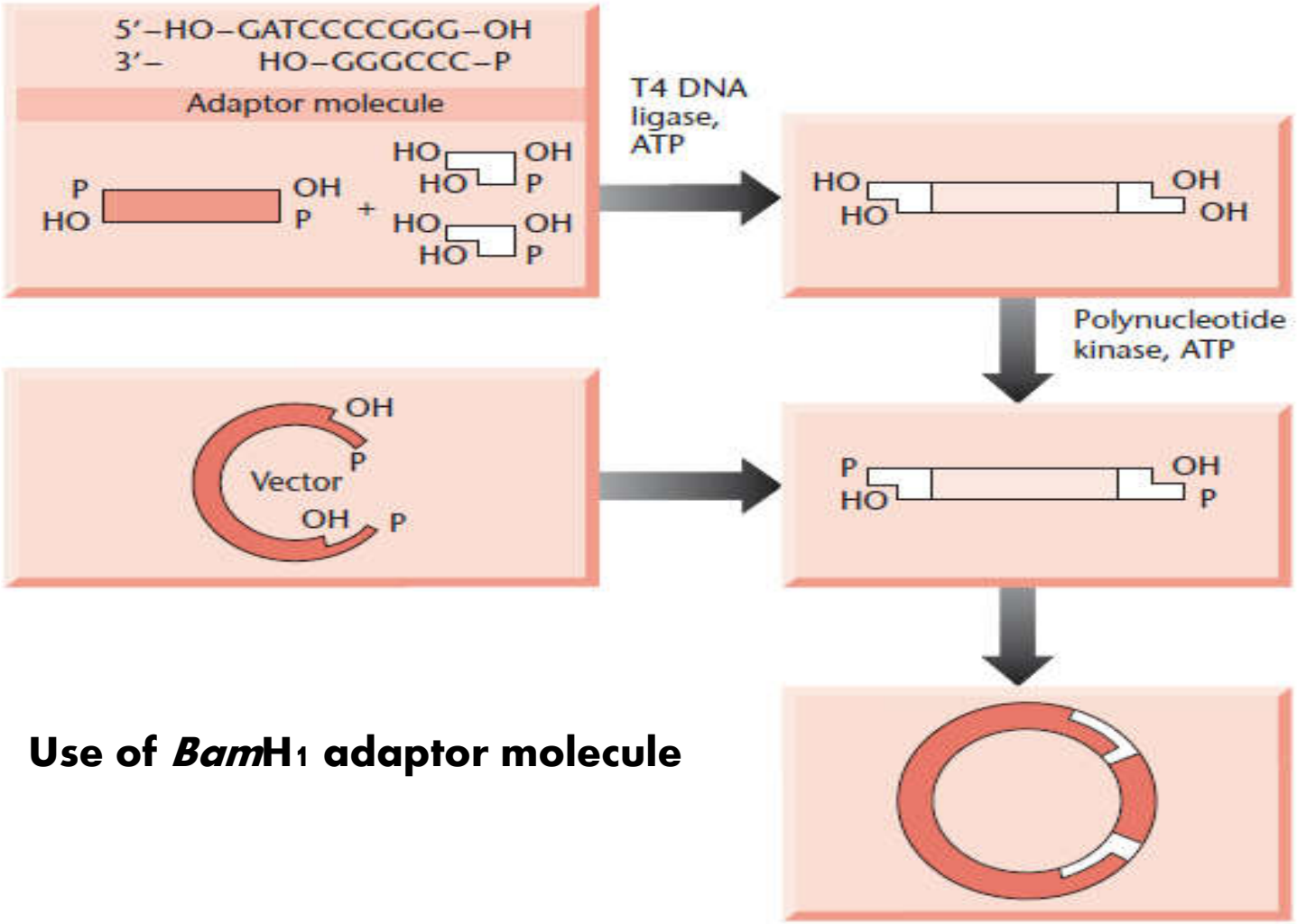
by T4 DNA ligase



Lecture #45-Joining DNA molecules

Adaptors

- **Chemically synthesized adaptor molecules which have a preformed cohesive end can be used to insert blunt ended foreign DNA into vector**



Use of *Bam*H1 adaptor molecule

Joining DNA molecules

Adaptors

- **The main difference between linker and adaptor is that former having blunt ends while later have one cohesive end**

Lecture # 46- Essentials of Genetics

SOUTHERN BLOTTING

- The Southern blot is used to detect the presence of a particular piece of DNA in a sample by a molecular probe.

SOUTHERN BLOTTING

- Southern Blotting is named after its inventor, the British Biologist Edwin Southern (1975).

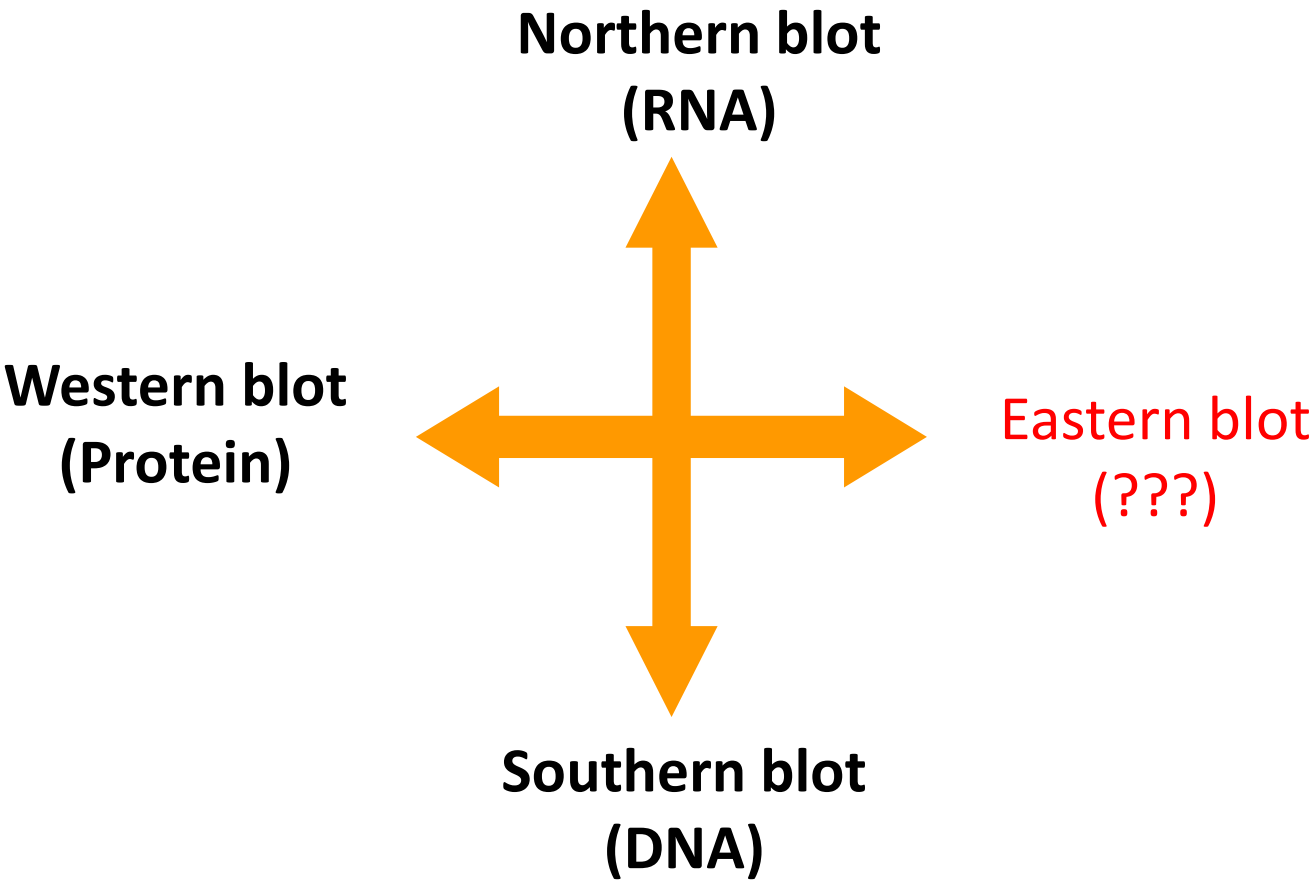
Southern Blotting

OTHER BLOTTING METHODS

- Other blotting methods with similar principles, but using protein or RNA, have been named in reference to Edwin Southern's name.

Southern Blotting

BACKGROUND



Southern Blotting

SOUTHERN BLOTTING

- Identify DNA sequence (gene) of interest.
- Identified DNA may be a small piece of DNA or a mutation.

Lecture # 47- Essentials of Genetics

SOUTHERN BLOTTING PROCEDURE

Southern Blotting - Procedure

PROCEDURE

- DNA is extracted from cells, leukocytes.
- DNA is cleaved into many fragments by restriction enzyme (e.g, BamH1, EcoR1 etc)
- The resulting fragments are separated on the basis of size by electrophoresis.
- The DNA fragments are denatured and transferred to nitrocellulose membrane for analysis.

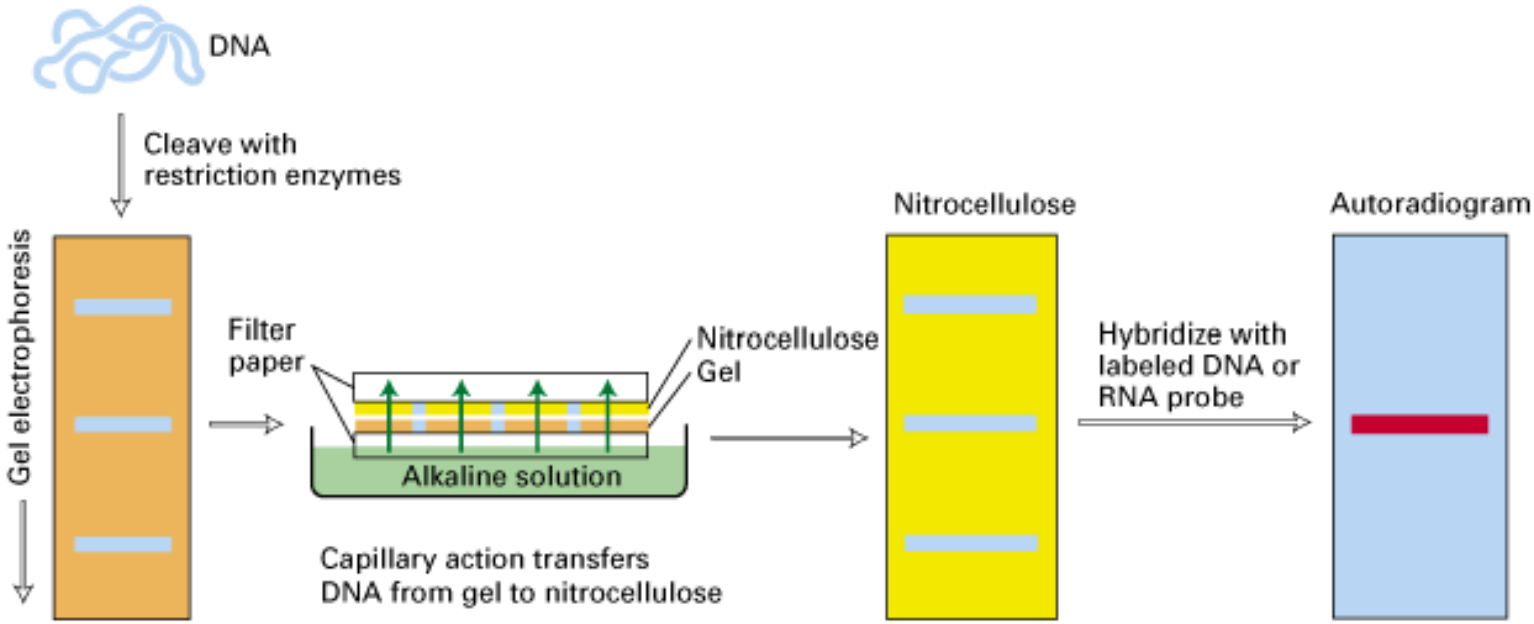
Southern Blotting - Procedure

PROCEDURE

- The labeled probe is added to the blocked membrane in buffer and incubated for several hours to allow the probe molecules to find their targets.
- Blot is incubated with wash buffers containing NaCl and detergent to wash away excess probe.
- Radioactive probes enable autoradiographic detection.

Southern Blotting - Procedure

PROCEDURE



Southern Blotting - Procedure

CONCLUSION

- Southern blot is used to detect the presence of a particular piece of DNA in a sample by a molecular probe.

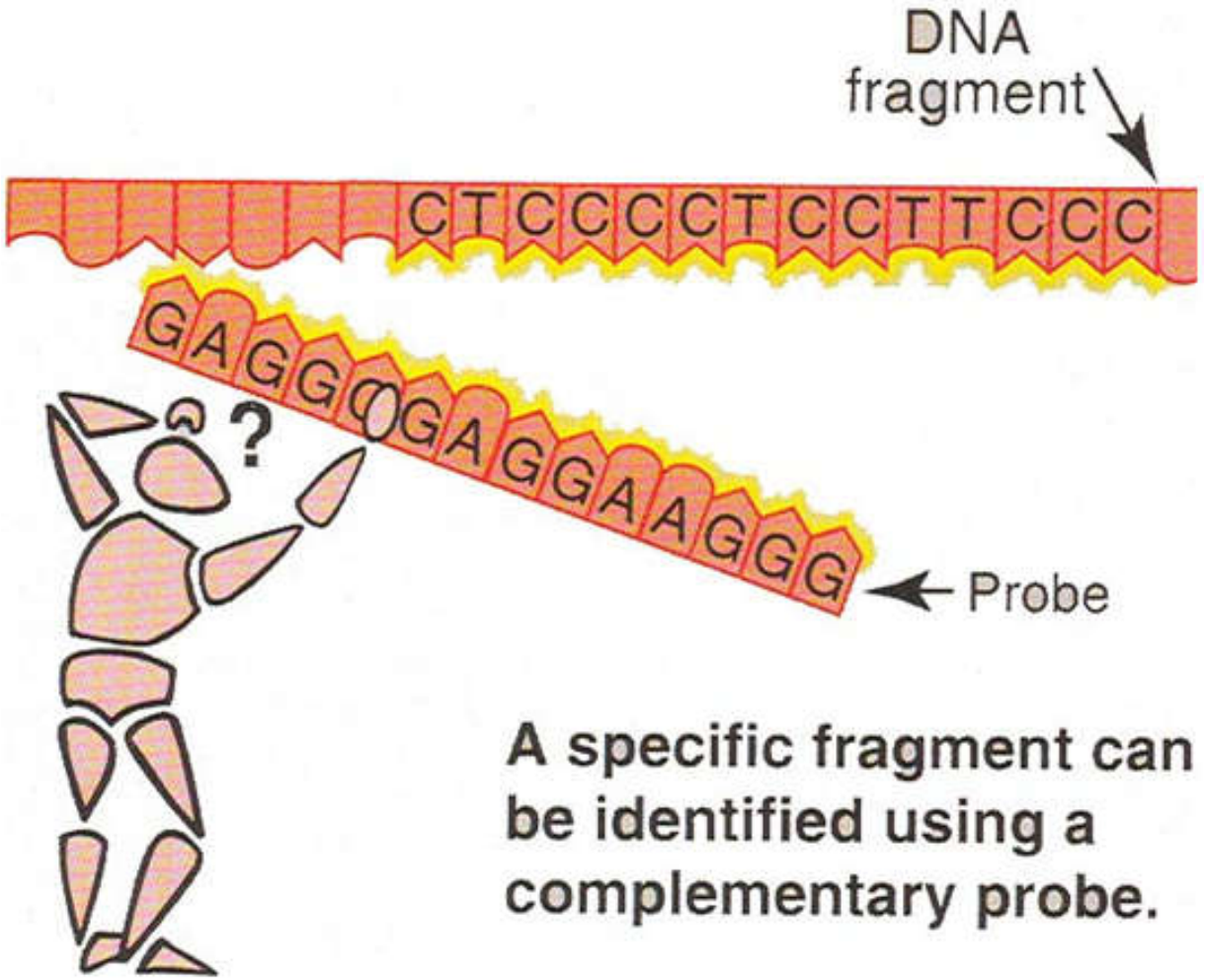
Lecture # 48- Essentials of Genetics

SOUTHERN BLOTTING PROBES

PROBES

- Labeled material to detect a target.
- For DNA: 20-30 nucleotides, complementary to a region in the gene or DNA.

Southern Blotting - Probes



A specific fragment can be identified using a complementary probe.

Southern Blotting - Probes

RADIOACTIVE PROBE - P32

- Sensitive
- Relatively cheap
- Hazardous
- Radioactive waste disposal regulations should be followed

NON-RADIOACTIVE PROBE - BIOTIN

- Sensitive
- Relatively expensive

Southern Blotting - Probes

HYBRIDIZATION OF PROBES

- The binding between single stranded labeled probe to a complementary nucleotide sequence on the target DNA.

PROBES

- Labeled material to detect complementary region in the gene or DNA.

Lecture # 49-Essentials of Genetics

TRANSFER METHODS IN SOUTHERN BLOTTING

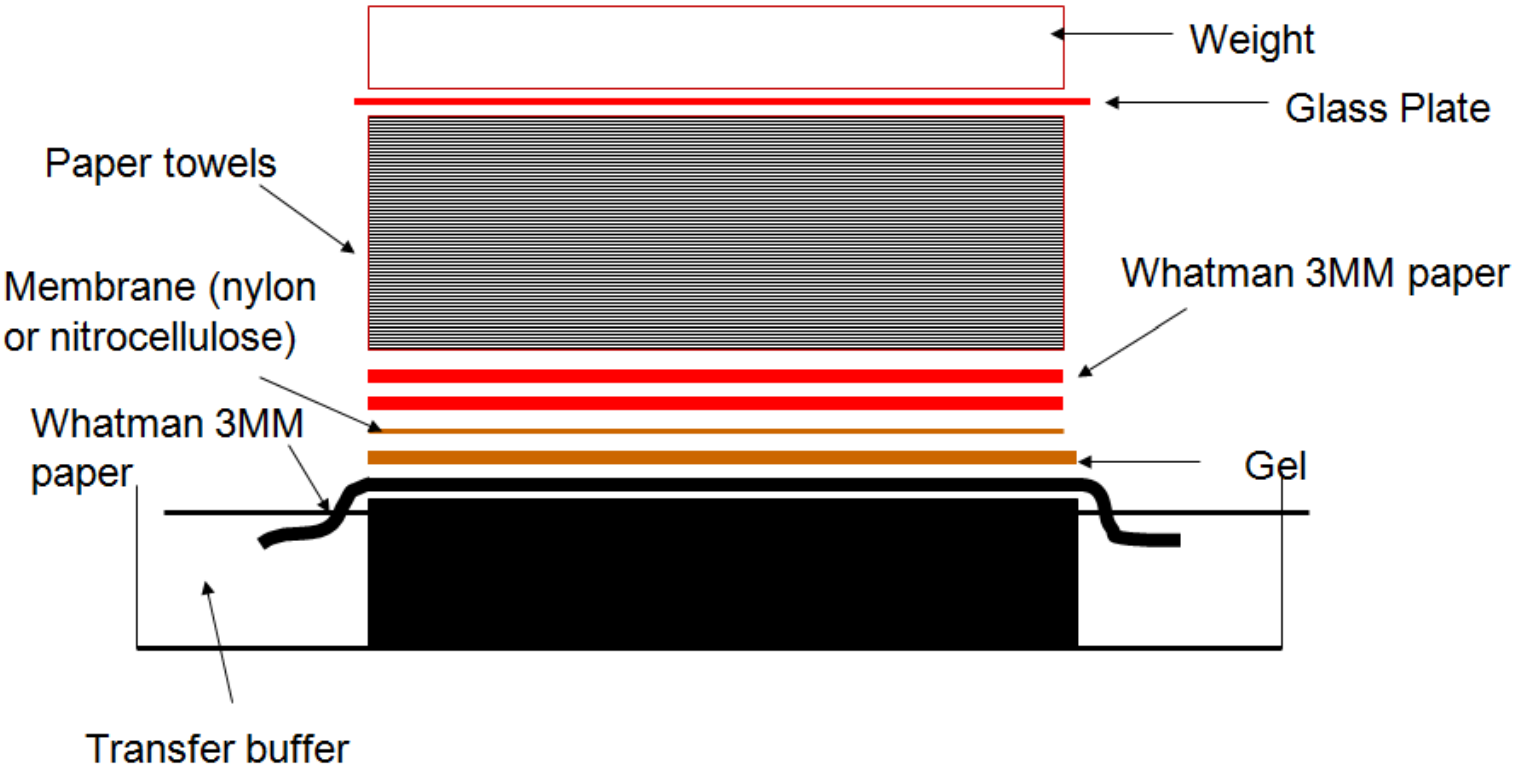
Transfer Methods - Southern

TRANSFER METHODS OF DNA TO MEMBRANE

- Upward capillary transfer
- Downward capillary transfer
- Simultaneous transfer to two membranes
- Electrophoretic transfer
- Vacuum transfer

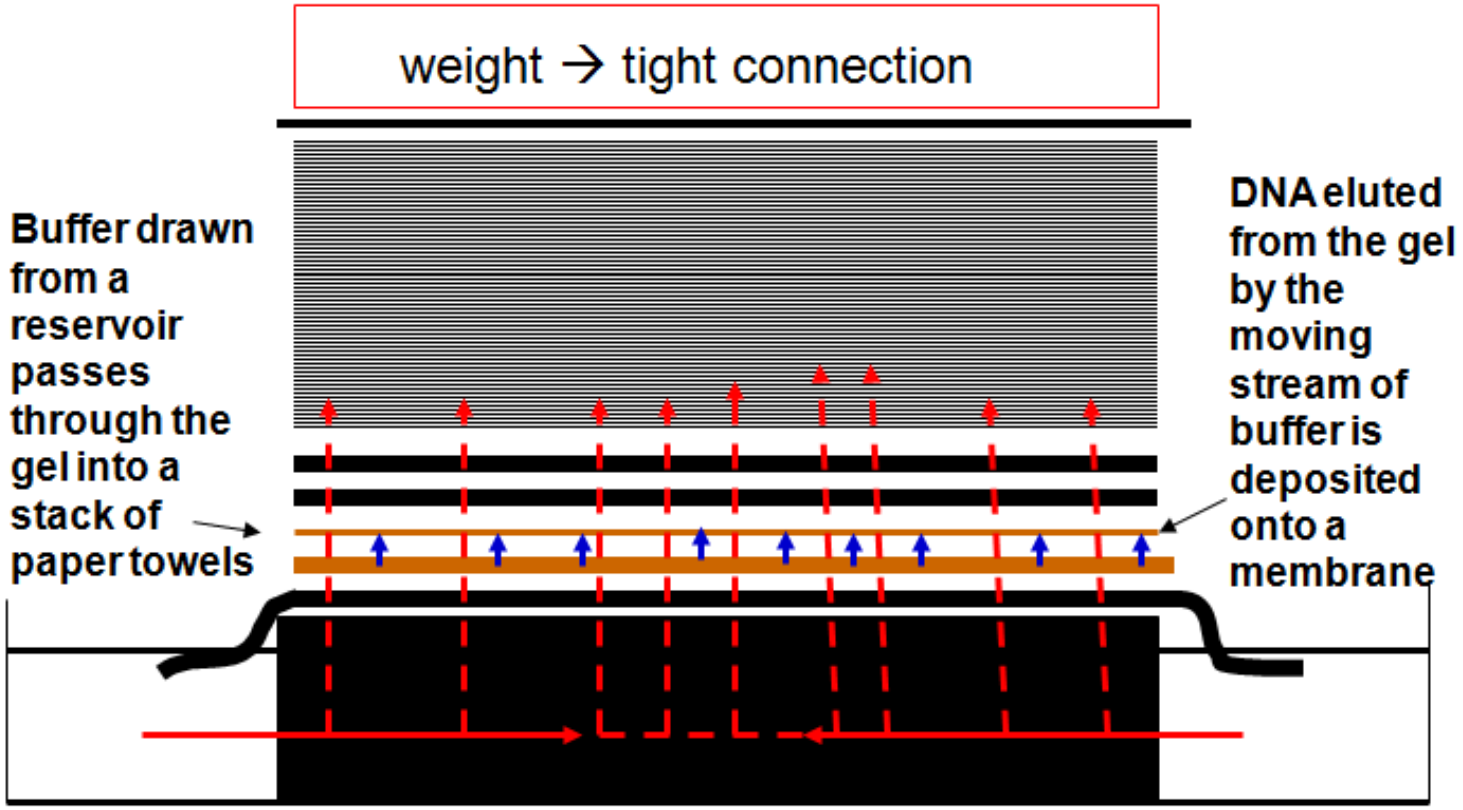
Transfer Methods - Southern

UPWARD TRANSFER



Transfer Methods - Southern

UPWARD TRANSFER OF DNA TO MEMBRANE



Transfer Methods - Southern

TRANSFER METHODS

- Upward capillary transfer
- Downward capillary transfer
- Simultaneous transfer to two membranes
- Electrophoretic transfer
- Vacuum transfer

Lecture # 50-Essentials of Genetics

MUTATIONS IDENTIFICATION BY SOUTHERN BLOTTING

MUTATION IDENTIFICATION

- The presence of a mutation affecting a restriction site causes the pattern of bands to differ from those seen in a normal gene.
- A change in one nucleotide may alter the nucleotide sequence so that the restriction endonuclease fails to recognize and cleave at that site.

Mutations Identification

MUTATION IDENTIFICATION

- Digestion of genomic DNA to DNA fragments.
- Size-separation of the fragments.
- In situ denaturation of the DNA fragments.
- Transfer of denatured DNA fragments into a solid support (nylon or nitrocellulose).
- Hybridization of the immobilized DNA to a labeled probe (DNA, RNA).
- Detection of the bands complementary to the probe (e.g. by autoradiography).

Mutations Identification

MUTATION IDENTIFICATION

- Estimation of the size & number of the bands generated after digestion of the genomic DNA will be different.

MUTATION IDENTIFICATION

- Mutations can be identified by Southern blotting.

Lecture # 51-Essentials of Genetics

NORTHERN BLOTTING

- Northern blot is a method used for detection of a specific RNA sequence in RNA samples.

PROCEDURE

- Isolation of intact mRNA.
- Separation of RNA according to size (through a denaturing agarose gel.
- Transfer of the RNA to a solid support.
- Fixation of the RNA.
- Hybridization of the immobilized RNA to probes complementary to the sequences of interest.

Northern Blotting

PROCEDURE

- Removal of probe molecules that are nonspecifically bound to the solid matrix.
- Detection, capture and analysis of an image of the specifically bound probe molecules.

NORTHERN BLOTTING

- Northern blot is a method used for detection of a specific RNA sequence in RNA samples.

52- Essentials of Genetics

NORTHERN BLOT APPLICATIONS

APPLICATIONS

- Study of gene expression in eukaryotic cells.
- To measure the amount & size of RNAs transcribed from eukaryotic genes.
- To estimate the abundance of RNAs.

Northern Blot - Applications

APPLICATIONS

- To equalize the amounts of RNA loaded into lanes of gels.
- Use of housekeeping gene (endogenous constitutively-expressed gene).
- Normalizing samples according to their content of mRNAs of the housekeeping gene.

- Northern blot is used in many ways while studying RNAs.

Lecture # 53-Essentials of Genetics

WESTERN BLOTTING

- A technique used to detect the presence of a specific protein in a complex protein mixture.

- A technique used to detect the presence of a specific protein in a complex protein mixture.

WESTERN BLOTTING

- To determine the molecular weight of a protein.
- To measure relative amounts (quantitation) of the protein present in complex mixtures of proteins that are not radiolabeled.

WESTERN BLOTTING

- Western blots have become one of the most common analytical tools for the detection of viral proteins.
- Characterization of monoclonal and polyclonal antibody preparations and in determining the specificity of the immune response to viral antigens.

Lecture # 54-Essentials of Genetics

WESTERN BLOTTING PROCEDURE

PROCEDURE

1- Sample preparation

2- Gel Electrophoresis

3- Blotting (or transfer)

4- Blocking

5- Antibody probing

6- Detection

Western Blotting - Procedure

PROCEDURE

1- The choice of extraction method depends primarily on the sample and whether the analysis is targeting all the proteins in a cell or only a component from a particular subcellular fraction.

2- Samples are loaded into separate wells. A protein marker is also loaded. The separated protein mixtures are transferred to a solid support for further analysis.

Western Blotting - Procedure

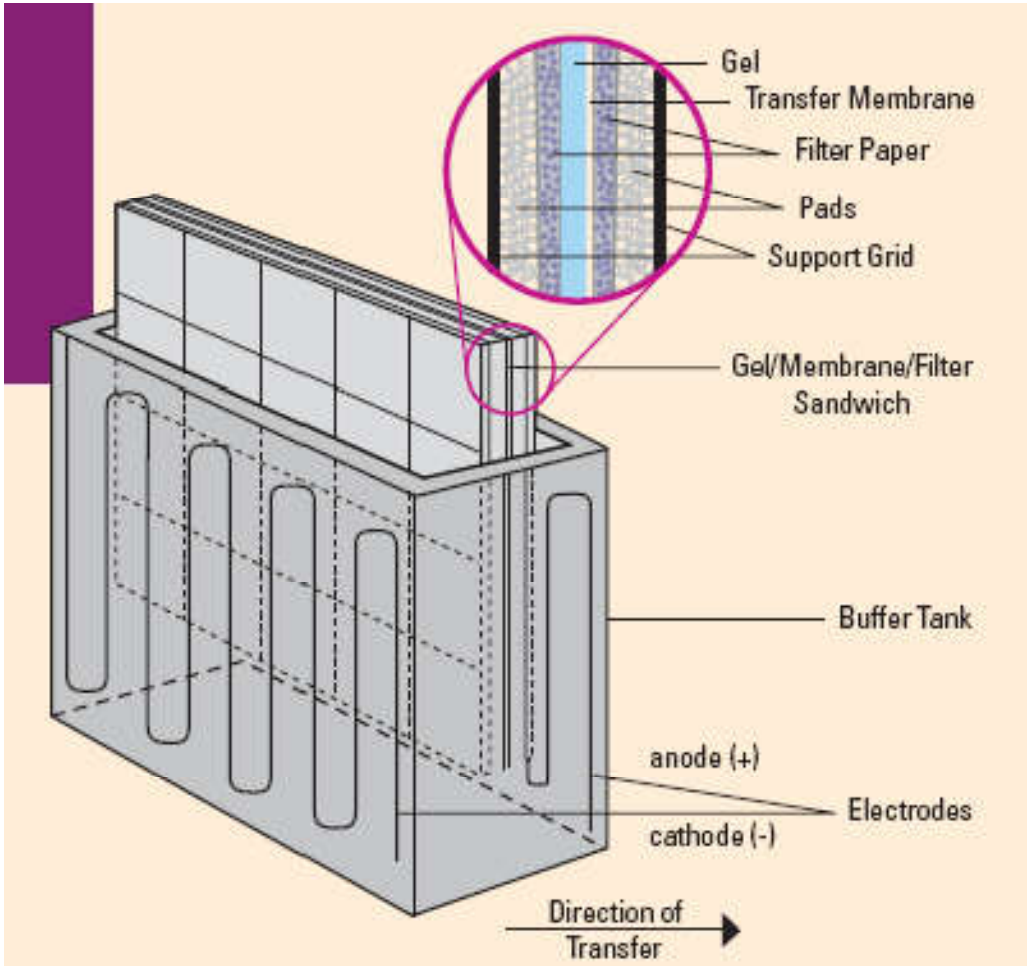
PROCEDURE

3- Transfer can be done in wet or semi-dry conditions. Semi-dry transfer is generally faster. Wet transfer is recommended for large proteins, >100 kD.

4- Blocking is a very important step in the immunodetection phase of Western blotting because it prevents non-specific binding of antibody to the blotting membrane.

Western Blotting - Procedure

ELECTROPHORETIC TRANSFER



Western Blotting - Procedure

PROCEDURE

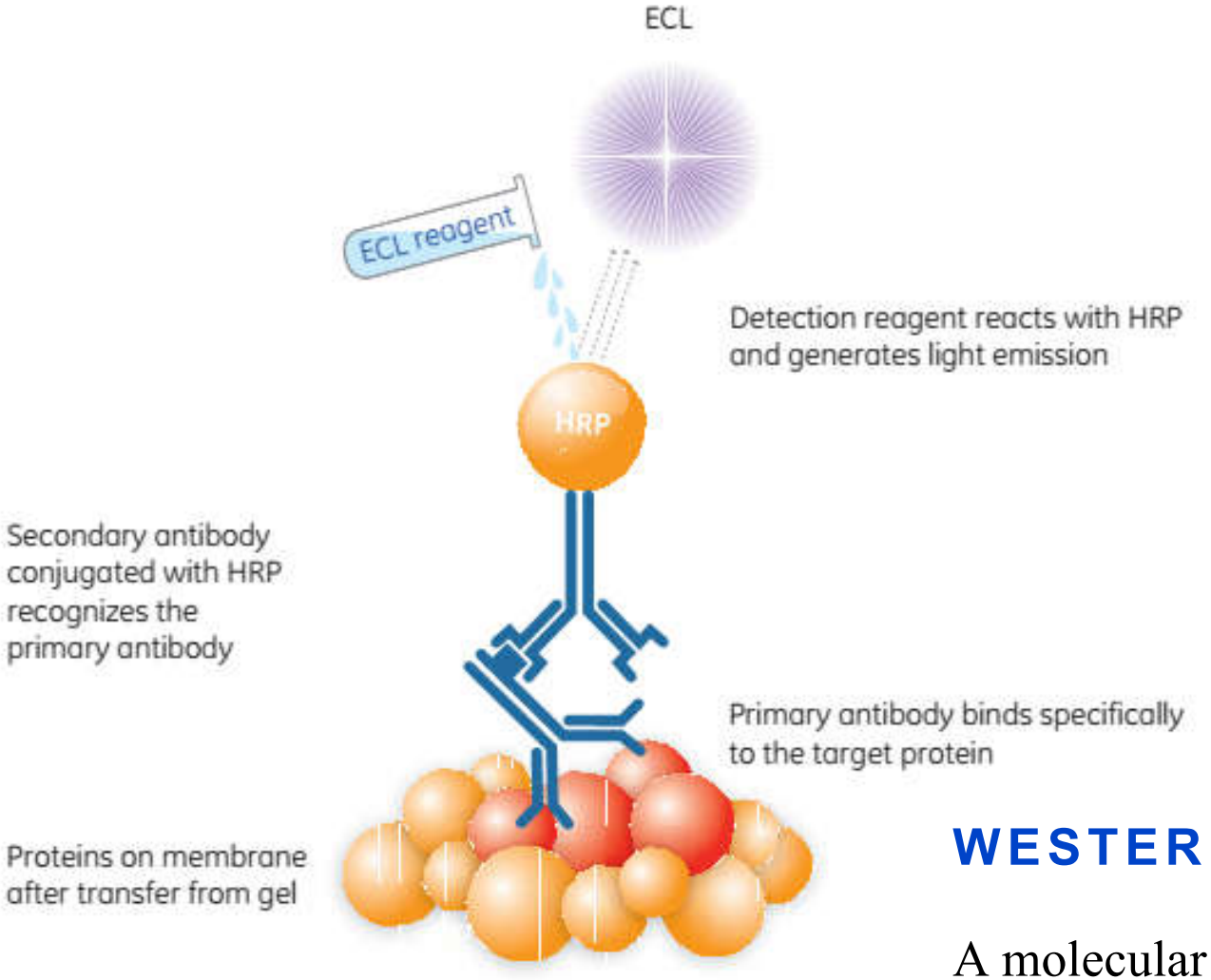
Protein of interest is detected and localized using a specific antibody. Western blotting protocols utilize a non-labeled primary antibody directed against the target protein.

A species-specific, labeled secondary antibody directed against the constant region of the primary antibody is then used. The most common antibody label used in Western blots is HRP.

PROCEDURE

The signal is detected when HRP is exposed to a substrate solution in the final step of the immunodetection procedure

Western Blotting - Procedure



WESTERN BLOTTING

A molecular technique to study proteins.